

Cover Page

Topic: Customer Churn Analysis

Exploratory Data Analysis (EDA) Report

Prepared By: Salman Rafi

Tools & Technologies: Python | Pandas | NumPy | Matplotlib | Seaborn | Jupyter
Notebook

Project Type: Data Analytics & Exploratory Data Analysis

Year: 01/03/2026

Table of Contents

1. Introduction
2. Project Objective
3. Dataset Overview
4. Data Cleaning & Preprocessing
5. Exploratory Data Analysis
6. Key Insights & Findings
7. Business Recommendations
8. Conclusion
9. Future Improvements

1. Introduction:

Customer churn is one of the major challenges faced by subscription-based businesses and service providers. Churn occurs when customers discontinue using a company's products or services. High churn rates negatively affect business revenue, customer acquisition costs, and long-term growth. This project performs Exploratory Data Analysis (EDA) on a Customer Churn dataset to identify important patterns, customer behaviors, and business factors associated with customer attrition. The analysis focuses on understanding how different variables such as contract type, payment methods, customer tenure, internet services, and demographic information influence churn behavior. The findings from this analysis can help organizations improve customer retention strategies and make more data-driven business decisions.

2. Project Objective:

The primary objective of this project is to analyze customer churn behavior and identify the key factors contributing to customer attrition.

Specific objectives include:

- Understanding customer churn distribution.
- Identifying high-risk customer groups.
- Analyzing the impact of contract types on churn.
- Evaluating the effect of payment methods on customer retention.
- Studying the relationship between customer tenure and churn.
- Generating business insights and actionable recommendations.
- Supporting customer retention strategy development.

3. Dataset Overview:

The dataset contains customer demographic information, service subscription details, account information, payment methods, and churn status.

Dataset Information:

- Total Rows: 7043
- Total Columns: 21
- Target Variable: Churn (Yes/No)

Important Features:

- Gender
- Senior Citizen Status
- Partner & Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

The dataset provides valuable information for analyzing customer behavior and identifying patterns related to customer retention and attrition.

4. Data Cleaning & Preprocessing:

Before performing the analysis, the dataset was cleaned and preprocessed to improve data quality and ensure accurate analysis.

Data Cleaning Steps:

- Checked for missing values
- Removed unnecessary columns
- Converted categorical variables into suitable formats
- Handled incorrect or inconsistent data entries
- Verified data types for numerical and categorical features
- Performed basic feature exploration

The preprocessing stage ensured that the dataset was clean, consistent, and suitable for exploratory data analysis.

5. Exploratory Data Analysis (EDA)

5.1 Churn Distribution Analysis:

The initial analysis focused on understanding the overall distribution of customer churn.

Findings:

- A significant portion of customers remained loyal to the company.
- However, a considerable percentage of customers discontinued services, indicating notable churn behavior.

Observation:

The churn distribution analysis highlights the importance of understanding customer dissatisfaction and improving retention strategies.

5.2 Contract Type Analysis:

Contract type emerged as one of the strongest indicators of customer churn.

Findings:

- Customers with month-to-month contracts showed the highest churn rate at approximately 42%.
- Customers with one-year contracts had a churn rate of nearly 11%.
- Customers with two-year contracts showed the lowest churn rate at approximately 3%.

Interpretation:

Customers with short-term contracts are more likely to discontinue services due to greater flexibility and lower commitment levels.

Business Impact:

High churn among short-term contract customers can increase customer acquisition costs and reduce long-term revenue stability.

5.3 Payment Method Analysis:

Payment method analysis revealed a strong relationship between payment behavior and customer churn.

Findings:

- Customers using electronic checks experienced the highest churn rate at approximately 45%.
- Customers using credit cards, bank transfers, or mailed checks showed significantly lower churn rates averaging between 15% and 18%.

Interpretation:

Electronic check users may experience payment inconvenience, trust issues, or weaker engagement with automated payment systems.

Business Impact:

Payment friction may negatively influence customer experience and contribute to customer attrition.

5.4 Customer Tenure Analysis:

Customer tenure analysis demonstrated a strong inverse relationship between tenure and churn **probability.**

Findings:

- Customers with tenure less than one year exhibited churn rates close to 50%.
- Customers with tenure between one and three years showed churn rates around 35%.
- Customers with tenure greater than three years had churn rates reduced to approximately 15%.

Interpretation:

New customers are significantly more vulnerable to churn during the early stages of their customer journey.

Business Impact:

Improving onboarding experience and customer engagement during the first year could significantly improve retention rates.

5.5 Internet Service Analysis:

The analysis also explored customer churn behavior based on internet service type.

Findings:

- Customers using Fiber Optic services showed churn rates around 30%.
- Customers using DSL services demonstrated comparatively lower churn rates near 20%.

Interpretation:

Higher churn among Fiber Optic customers may indicate pricing concerns, competition, or dissatisfaction with service quality.

Business Impact:

Service quality improvements and enhanced customer support may help reduce churn among Fiber Optic users.

5.6 Senior Citizen Analysis:

Customer demographic analysis identified notable differences in churn behavior between senior citizens and non-senior customers.

Findings:

- Senior citizens experienced churn rates around 41%.
- Non-senior customers showed lower churn rates near 26%.

Interpretation:

Senior customers may require simplified services, personalized assistance, and improved customer support experiences.

Business Impact:

Targeted customer engagement programs for senior customers may improve retention and customer satisfaction.

6. Key Insights & Findings:

Major findings from the analysis include:

- Month-to-month contract customers demonstrated the highest churn behavior.
- Electronic check payment users showed significantly higher churn rates compared to other payment methods.
- Customer churn probability decreased as tenure increased.
- Fiber Optic customers exhibited higher churn rates than DSL users.
- Senior citizens showed relatively higher churn tendencies.
- Monthly charges and short-term contracts strongly influenced customer attrition.

The analysis highlights that customer retention is strongly associated with customer commitment level, payment convenience, and service experience.

7. Business Recommendations:

7.1 Promote Long-Term Contracts:

- Provide discounts, loyalty rewards, and bundled offers to encourage customers to switch from month-to-month contracts to yearly or bi-annual plans.

7.2 Improve Early Customer Engagement:

- Since customers within the first year show the highest churn probability, businesses should improve onboarding experiences, customer support, and engagement during the early customer lifecycle.

7.3 Optimize Payment Experience:

- Encourage customers to adopt automated payment methods such as credit cards and bank transfers by offering incentives and simplifying payment processes.

7.4 Improve Fiber Optic Customer Satisfaction:

- Conduct customer satisfaction surveys and improve service reliability for Fiber Optic customers to reduce churn rates.

7.5 Develop Personalized Retention Strategies:

Create targeted retention programs for high-risk customer groups including:

- Senior citizens
- Short-term contract customers
- High monthly charge customers

8. Conclusion:

The analysis successfully identified several critical factors influencing customer churn behavior.

Contract type, payment method, customer tenure, internet service type, and demographic characteristics all demonstrated strong relationships with customer retention and attrition.

Customers with short-term contracts, electronic check payment methods, and shorter tenure periods were significantly more likely to churn.

The findings suggest that businesses can substantially improve customer retention through:

- Long-term contract incentives
- Improved onboarding experiences
- Better payment systems
- Personalized customer engagement strategies

This project demonstrates how exploratory data analysis can provide meaningful business insights and support data-driven decision-making processes.

9. Future Improvements:

Future enhancements for this project may include:

- Customer churn prediction using Machine Learning models
- Interactive Power BI or Tableau dashboard integration
- Real-time customer churn monitoring systems
- Customer segmentation analysis
- Sentiment analysis using customer feedback data
- SQL database integration for scalable analytics